FAIRDOM-SEEK: Platform for FAIR data and research outcome management

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FAIRDOM SEEK

The **FAIRDOM-SEEK** (fairdomseek.org) Platform is a customisable open-source, web-based catalogue designed to facilitate FAIR research data management, enabling the storage, cataloging, sharing, and reuse of research outcomes. Developed and maintained by the **FAIRDOM consortium** (fair-dom.org), FAIRDOM-SEEK **empowers scientists** to organise, document, and publish their research data, utilizing the ISA framework to structure experiments and related assets like protocols. **Key features** include robust data, sample and metadata management, version control, linking to other repositories and integration with modeling tools. **Collaboration** is supported through controlled sharing, while DOI creation enables long-term data accessibility.

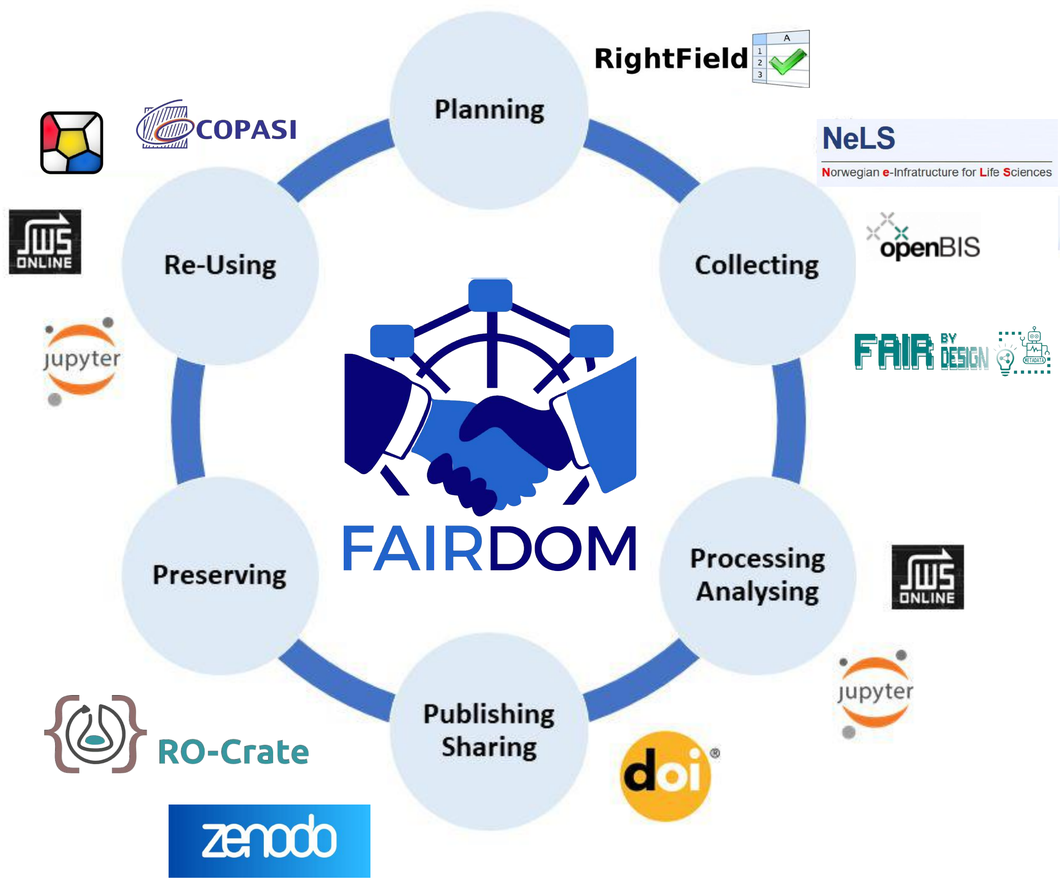
Customised Installations

FAIRDOM-SEEK allows customisation, and encourages contributions, leveraging the platform's core functionalities in distinct ways, tailoring its appearance, structure, and content.

Notable examples include:

- FAIRDOM HUB — fairdomhub.org
- WorkflowHub — workflowhub.eu
- DATA HUB — datahub.elixir-belgium.org
- nfdi 4 health — www.nfdi4health.de/en/service/local-data-hub
- IBISBA — hub.ibisba.eu
- MIT BioMicro Center — openwetware.org/wiki/BioMicroCenter

FAIRDOM support for Research Data Lifecycle

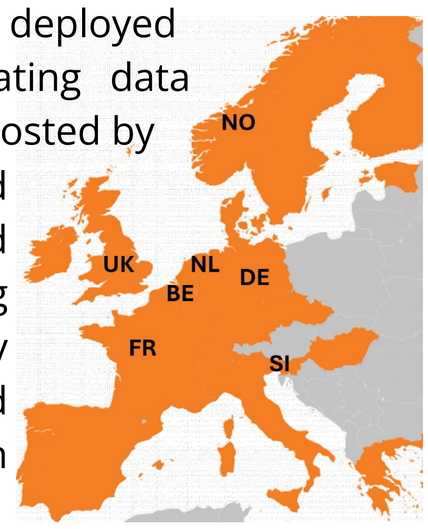


Recent key improvements

- Enhanced integration with the Norwegian e-Infrastructure for Life Sciences, **NeLS** (elixir.no/Services-bak/nels)
- Research Organization Registry (**ROR**) annotation support
- **Extended metadata** framework
- Integration with **FAIR Data Station** (fairds.fairbydesign.nl) in conjunction with the **BioIndustry4.0** (www.bioindustry4.eu) and **PhenomUK** (phenomuk.org) projects
- **Copasi** and **Morpheus** integration for modelling

Usage across ELIXIR Nodes

Many instances, or "**Hubs**," have been deployed internationally by scientific consortia, facilitating data storage, sharing, and analysis. **FAIRDOMHub**, hosted by ELIXIR Germany, serves over 450 national and international projects as both a repository and knowledge-sharing platform, promoting interdisciplinary collaboration and community building. As a core component of de.NBI and ELIXIR, it significantly contributes to European life science data management.



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